



DIGITAL TECHNOLOGIES

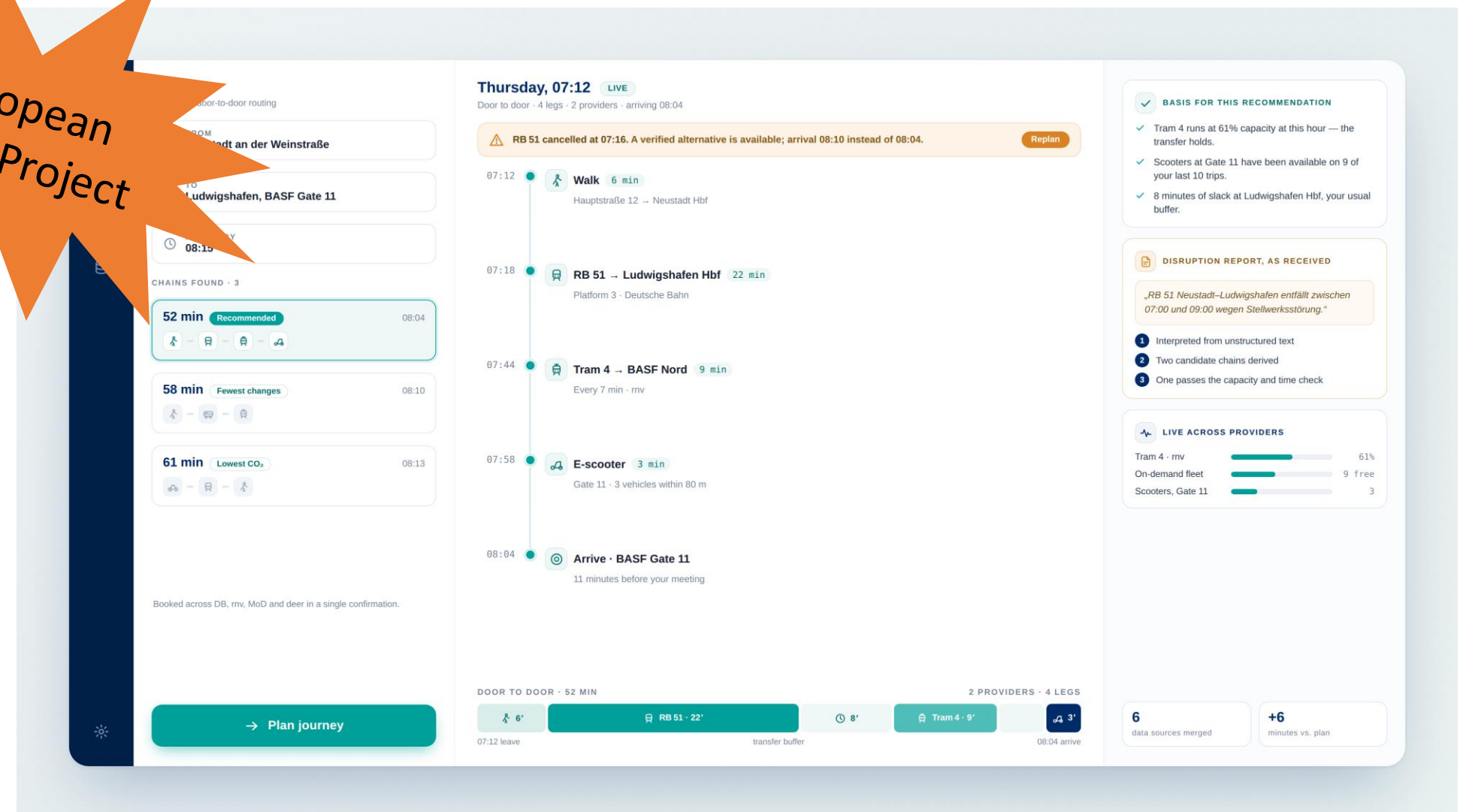
ZUKUNFT. DIGITAL. NACHHALTIG.

WiSe 2026

FastMile

Unified Mobility Platform for LLM-Guided Routing

Machine Learning and COgnitive SoftwaRE (CORE)
Research Group
AI Team Projects (ATP)



Ein gemeinsames Studienprogramm der

THE STARTING POINT

Daily commuting, across every mode that fits

A polycentric region

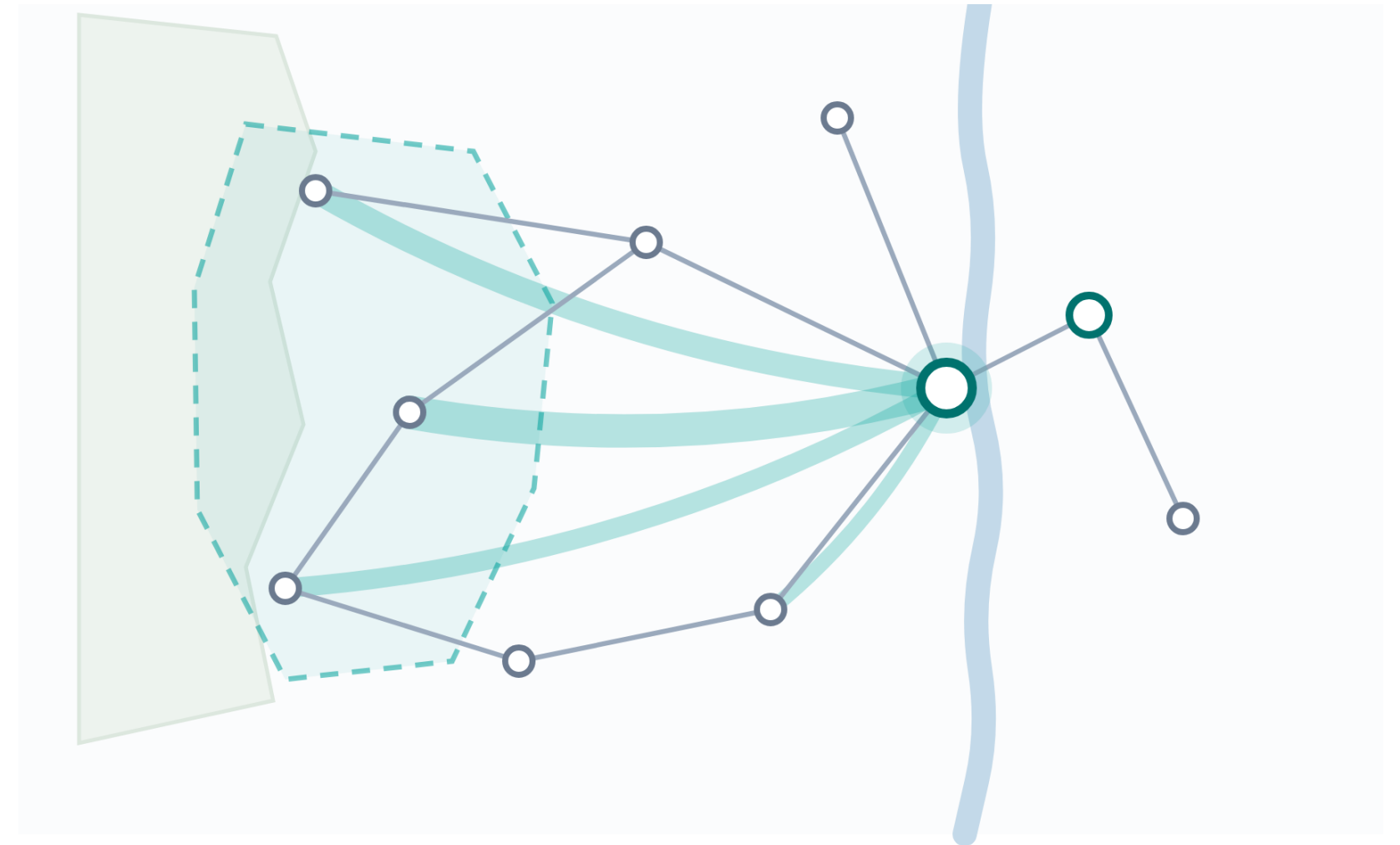
Commuters travel from surrounding towns and districts into larger employment centres

Several modes, one journey

Public transport, on-demand services, car sharing and micromobility are combined into a single door-to-door chain

The whole chain decides

The private car still dominates because it is more often the option that works reliably from the front door to the destination

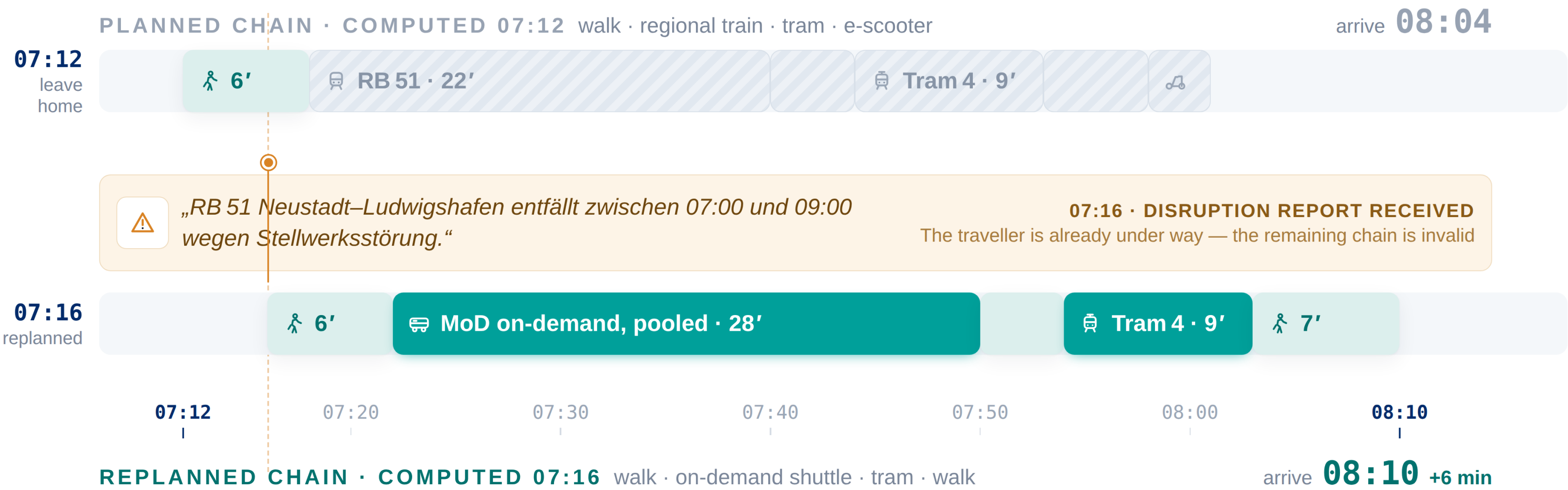


OBJECTIVE

Provide convenient and predictable transfer options across the whole intermodal journey

INCIDENT HANDLING

Disruptions need to be adressed on the go



Detection
The report is read from unstructured text

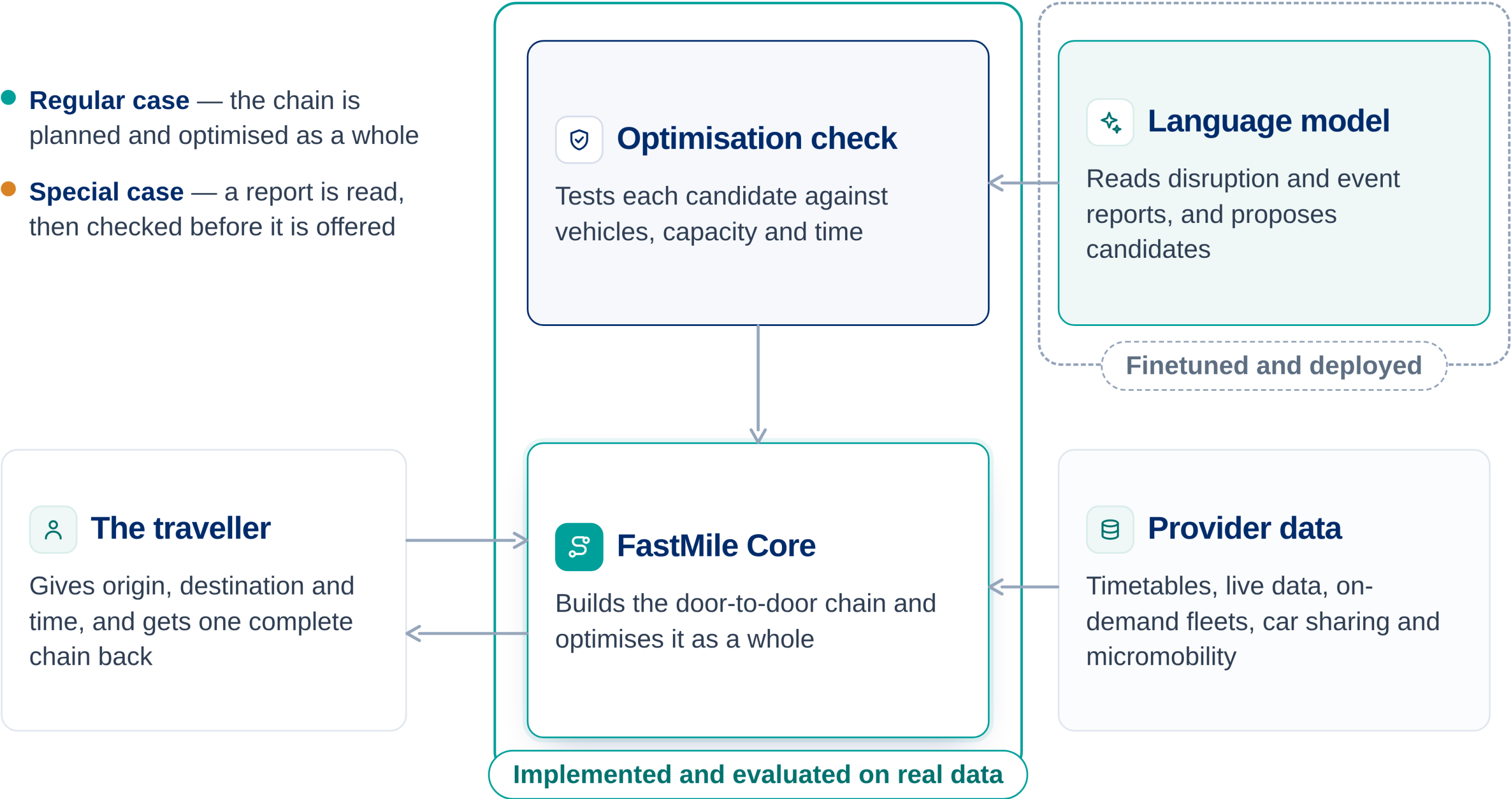
Replanning
Candidate chains are derived

Verification
Checked against capacity and remaining time

SYSTEM ARCHITECTURE

What you will implement

- **Regular case** — the chain is planned and optimised as a whole
- **Special case** — a report is read, then checked before it is offered



DATA BASIS

Data sources

SOURCE

Public transport timetables, stops and real-time data

Utilisation of lines and vehicles

Trip data from on-demand services

Car-sharing and micromobility offers

Disruption reports and event announcements

USED FOR

Intermodal connections that work in practice

Capacity-aware recommendations, including at peak times

Fleet state, pooling opportunities, demand patterns

Further modes available inside a single chain

The trigger for the special cases, arriving as free text

We use real usage data – not synthetic data

DEV-TEAM

Thinking about joining?

WHAT YOU BRING

- Comfortable writing **Python**
- Comfortable with **git** and a shared repository
- Happy working in an international team across two universities
- Interested in optimization and machine learning on live data

WHAT YOU LEARN

- Interpreting unstructured operational text with language models
- Combining machine learning with analytical route optimisation
- Building a web application on live data from several providers
- Taking a research question through to a system that runs

Be part of an international team!

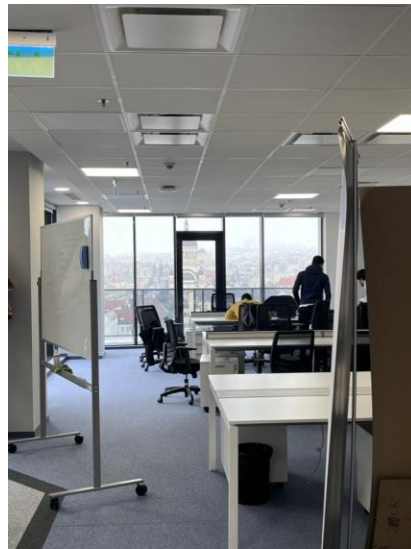
together with students from Babeş-Bolyai University, Cluj, Romania

For more info
and videos
visit:



<https://www.core-network.ai/ai-team-projects>

Intensive Week in Cluj



Social Events



SAVE THE DATE

Intensive Week Cluj:

~ 11.11. – 2.11.2026

Costs for flight and accommodation will be covered

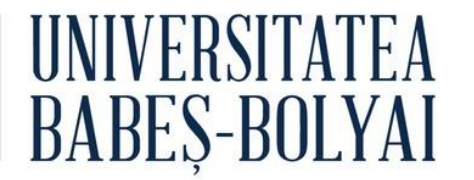
Intensive Week Goslar:

~ 01.02. – 07.02.2027

CORE Projects in WiSe26

Kairos

FastMile



Machine Learning and COgnitive SoftwaRE (CORE)

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